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System for mobile robot to autonomously take elevator

Abstract

A system for a mobile robot to autonomously take an elevator includes an elevator button pressing module mounted on an elevator floor control panel of the elevator. The elevator button pressing module communicates with the mobile robot wirelessly and is adapted to press a pair of elevator buttons on the elevator floor control panel according to an instruction issued by the mobile robot to call an elevator car.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the benefit of the filing date under 35 U.S.C. § 119(a)-(d) of Chinese Patent Application No. 201910197908.9, filed on Mar. 15, 2019.

FIELD OF THE INVENTION

(2) The present invention relates to an elevator and, more particularly, to a system for a mobile robot to autonomously or automatically take an elevator.

BACKGROUND

(3) In order to ensure the safety of elevators, third parties are generally not allowed to make any modifications to the elevators. Therefore, in the prior art, a mobile robot cannot control the elevator by communicating with the elevator control system. The mobile robot, consequently, can only move on a single floor and cannot move to another floor by autonomously taking the elevator; the elevator must be manually operated by a worker to transport the mobile robot to another floor if there is a need to move the mobile robot to another floor, limiting the application of mobile robots.

SUMMARY

(4) A system for a mobile robot to autonomously take an elevator includes an elevator button pressing module mounted on an elevator floor control panel of the elevator. The elevator button pressing module communicates with the mobile robot wirelessly and is adapted to press a pair of elevator buttons on the elevator floor control panel according to an instruction issued by the mobile robot to call an elevator car.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will now be described by way of example with reference to the accompanying Figures, of which:

(2) FIG. 1 is a front perspective view of an elevator button pressing module according to an embodiment;

(3) FIG. 2 is a rear perspective view of the elevator button pressing module;

(4) FIG. 3 is a front view of an elevator floor door and an elevator floor control panel according to an embodiment;

(5) FIG. 4 is a perspective view of the elevator button pressing module mounted on the elevator floor control panel; and

(6) FIG. 5 is a perspective view of a mobile robot using the elevator button pressing module.

DETAILED DESCRIPTION OF THE EMBODIMENT(S)

(7) The technical solution of the disclosure will be described in detail in the following embodiments with reference to the accompanying drawings. In the specification, the same or similar reference numerals denote the same or similar components. The following description of the embodiments of the present disclosure with reference to the accompanying drawings is intended to explain the general inventive concept of the disclosure and should not be construed as a limitation of the present disclosure.

(8) In addition, in the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the embodiments of the disclosure. However, one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are shown in schematic form in order to simplify the drawing.

(9) A system for a mobile robot 1 to autonomously take an elevator, as shown in FIGS. 1-5,

comprises an elevator button pressing module **10**. The elevator button pressing module **10** is mounted on an elevator floor control panel **20** of the elevator. The elevator button pressing module **10** communicates with the mobile robot **1** wirelessly, and is adapted to press corresponding elevator buttons **21**, **22** on the elevator floor control panel **20** according to an instruction issued by the mobile robot **1** to call an elevator car.

(10) The elevator button pressing module **10**, as shown in FIGS. **1** and **2**, includes a housing **100**, a pair of virtual fingers **111**, **112**, a driver **113**, and a controller **114**.

(11) As shown in FIGS. **1-5**, the housing **100** is adapted to be mounted on the elevator floor control panel **20**, and has a back side **120** facing the elevator floor control panel **20**, and a front side **110** facing away from the elevator floor control panel **20**.

(12) The virtual fingers **111**, **112**, as shown in FIG. **2**, are provided on the back side **120** of the housing **100** and are configured to be movable relative to the housing **100**. The driver **113** is provided in the housing **100** and is used for driving the virtual fingers **111**, **112** to move. The controller **114** is provided in the housing **100** and is used to control the driver **113**.

(13) The controller **114**, shown in FIG. **1**, is configured to communicate with the mobile robot **1** wirelessly and is adapted to control the driver **113** according to an instruction issued by the mobile robot **1**. The driver **113** is adapted to drive the virtual finger **111** or **112** to move to press a corresponding elevator button **21** or **22** on the elevator floor control panel **20** under the control of the controller **114**.

(14) The elevator button pressing module **10** further comprises virtual buttons **101**, **102**. The virtual buttons **101**, **102** are provided on the front side **110** of the housing **100**, as shown in FIGS. **1**, **4**, and **5**. The driver **113** is configured to, in response to the virtual button **101** or **102** being pressed, drive the virtual finger **111** or **112** to press the elevator button **21** or **22** corresponding to the pressed virtual button **101** or **102** on the elevator floor control panel **20** under the control of the controller **114**.

(15) The elevator button pressing module **10**, as shown in FIG. **1**, includes a display **103** provided on the front side **110** of the housing **100** and configured to display a working status of the elevator button pressing module **10** and/or the elevator.

(16) The elevator button pressing module **10** further comprises a data interface **105**, as shown in FIGS. **1** and **2**. The elevator button pressing module **10** is adapted to be connected to an external device, for example, to an external computer, a sensor, or the like, through the data interface **105**.

(17) The elevator button pressing module **10**, as shown in FIGS. **1** and **2**, includes a power switch **104**. The power switch **104** is provided on the top of the housing **100** and is used to control a power supply to the elevator button pressing module **10**.

(18) As shown in FIGS. **4** and **5**, the system further comprises a sensor **40**. The sensor **40** is provided near an elevator floor door **30**, connected to the controller **114** of the elevator button pressing module **10**, and configured for detecting whether or not the elevator floor door **30** is opened. The controller **114** of the elevator button pressing module **10** notifies the mobile robot **1** that it can enter the elevator car when the sensor **40** detects that the elevator floor door **30** has been opened. In various embodiments, the sensor **40** may be a non-contact distance sensor or an image sensor.

(19) The elevator buttons **21**, **22**, shown in FIG. **3**, include an elevator up button **21** and an elevator down button **22** provided on the elevator floor control panel **20**. The elevator button pressing module **10** is adapted to press the elevator up button **21** and the elevator down button **22** on the elevator floor control panel **20**.

(20) The virtual fingers **111**, **112**, as shown in FIG. **2**, include a first virtual finger **111** and a second virtual finger **112**. The first virtual finger **111** is used to press the elevator up button **21**, and the second virtual finger **112** is used to press the elevator down button **22**. The driver **113** drives the first virtual finger **111** to press the elevator up button **21** under the control of the controller **114** when the mobile robot **1** issues an instruction to press the elevator up button **21**. The driver **113**

drives the second virtual finger **112** to press the elevator down button **22** under the control of the controller **114** when the mobile robot **1** issues an instruction to press the elevator down button **22**. (21) The virtual buttons **101**, **102**, as shown in FIGS. **1**, **4**, and **5**, include a first virtual button **101** and a second virtual button **102**. The first virtual button **101** corresponds to the elevator up button **21**, and the second virtual button **102** corresponds to the elevator down button **22**. The driver **113** drives the first virtual finger **111** to press the elevator up button **21** under the control of the controller **114** when the first virtual button **101** is pressed. The driver **113** drives the second virtual finger **112** to press the elevator down button **22** under the control of the controller **114** when the second virtual button **102** is pressed.

(22) In the foregoing various exemplary embodiments according to the present disclosure, the mobile robot **1** can autonomously take the elevator using the system. Therefore, the present disclosure expands the application range of the robots and does not require any modification to the elevator itself, which will not have any impact on the safety of the elevator.

(23) Those skilled in the art will appreciate that the above-described embodiments are illustrative and can be modified, and that the structures described in the various embodiments can be freely combined without conflict in structure or principle. Although the present disclosure has been described with reference to the accompanying drawings, the embodiments disclosed in the drawings are intended to be illustrative explanation of the embodiments of the disclosure, and should not be construed as limiting the disclosure. Although several embodiments of the present general inventive concept have been shown and described, it would be appreciated by those skilled in the art that changes may be made in these embodiments without departing from the principles and spirit of the present general inventive concept, and the scope of the present disclosure is defined by the claims and their equivalents.

Claims

1. A system, comprising: a mobile robot; an elevator having an elevator floor control panel; and an elevator button pressing module mounted on the elevator floor control panel of the elevator, the elevator button pressing module communicates with the mobile robot wirelessly and is adapted to press a pair of elevator buttons on the elevator floor control panel according to an instruction issued by the mobile robot to call an elevator car, the elevator button pressing module includes: a pair of virtual fingers on a back side of a housing; a pair of virtual buttons on a front side of the housing; a driver in the housing driving the virtual fingers to move relative to the housing; and a controller disposed in the housing and controlling the driver, the controller wirelessly communicates with the mobile robot, the driver drives one of the virtual fingers to move to press a corresponding one of the elevator buttons on the elevator floor control panel in response to one of the virtual buttons being pressed and under the control of the controller according to the instruction issued by the mobile robot.
2. The system of claim 1, wherein the housing is mounted on the elevator floor control panel, the housing having the back side facing the elevator floor control panel and the front side facing away from the elevator floor control panel.
3. The system of claim 1, wherein the elevator button pressing module includes a display on the front side of the housing, the display configured to display a working status of the elevator button pressing module and/or the elevator.
4. The system of claim 1, wherein the elevator button pressing module has a data interface connected to an external device.
5. The system of claim 1, wherein the elevator button pressing module has a power switch on a top of the housing and controlling a power supply to the elevator button pressing module.
6. The system of claim 1, further comprising a sensor connected to the controller of the elevator button pressing module and detecting whether or not an elevator floor door is opened.

7. The system of claim 6, wherein the controller notifies the mobile robot that it can enter the elevator car in response to the sensor detecting that the elevator floor door has been opened.
 8. The system of claim 1, wherein the elevator buttons include an elevator up button and an elevator down button, the elevator button pressing module presses the elevator up button and the elevator down button.
 9. The system of claim 8, wherein the virtual fingers include a first virtual finger used to press the elevator up button and a second virtual finger used to press the elevator down button.
 10. The system of claim 9, wherein the driver drives the first virtual finger to press the elevator up button under the control of the controller in response to the mobile robot issuing an instruction to press the elevator up button.
 11. The system of claim 9, wherein the driver drives the second virtual finger to press the elevator down button under the control of the controller in response to the mobile robot issuing an instruction to press the elevator down button.
 12. The system of claim 8, wherein the virtual buttons include a first virtual button corresponding to the elevator up button and a second virtual button corresponding to the elevator down button.
 13. The system of claim 12, wherein the driver drives the first virtual finger to press the elevator up button under the control of the controller in response to the first virtual button being pressed.
 14. The system of claim 12, wherein the driver drives the second virtual finger to press the elevator down button under the control of the controller in response to the second virtual button being pressed.
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